MINOR PROJECT : AI Domain

Predictive Customer Churn Model for a Telecom Company

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Course: BE – Computer Science Engineering

Domain: Artificial Intelligence(AI)

Internship : Minor Project

Organisation: Persevex Internship

Year: 2026

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Problem Statement

- **Customer churn** refers to the **loss of customers** who stop using a company's services.
- In highly competitive industries such as **telecommunications, customer retention** is crucial for maintaining **revenue and business growth.**
- **Acquiring new customers** is significantly more expensive than **retaining existing ones.**
- Therefore, organisations need a **reliable system** to identify customers who are at risk of churning.
- This project aims to develop a **predictive model** that can accurately identify **potential churners**, enabling businesses to take proactive measures and improve **customer satisfaction.**
- Additionally, early churn prediction helps companies design personalized offers, improve customer engagement, and **enhance overall service quality.**

Objectives

- To perform **Exploratory Data Analysis (EDA)** to understand **customer churn patterns**
- To apply **Feature Engineering** for creating **meaningful predictive features**
- To preprocess data using **Encoding** and **Feature Scaling techniques**
- To build and compare **ML models (Logistic Regression, Random Forest, XGBoost)**
- To optimise models using **Hyperparameter Tuning and Cross-Validation**
- To evaluate performance using **AUC-ROC, Precision, Recall, F1-Score**
- To interpret models using **SHAP (Explainable AI)**
- To provide **actionable insights** for improving **customer retention**

Telco Customer Churn Dataset

- The dataset used in this project is the **Telco Customer Churn Dataset**, which contains comprehensive information about **customer demographics, services, and billing details.**

📊 **Features Included:**

- **Customer Information**: Gender, Senior Citizen, Partner, Dependents
- **Account Information**: Tenure (duration of customer relationship)
- **Contract Type** (Monthly, One-year, Two-year) , **Paperless Billing, Payment Method**
- **Services Subscribed**: Phone Service, Multiple Lines
- **Internet Service** (DSL, Fiber Optic, None), **Online Security, Online Backup, Device Protection, Tech Support , Streaming TV, Streaming Movies ,Tenure**
- **Billing Information**: Monthly Charges, Total Charges

🎯 **Target Variable:**

- Churn (Yes/No)
- This dataset enables **detailed analysis of customer behavior**, helping in **accurate churn prediction.**

Data Preprocessing

- Data preprocessing is a **critical step** to ensure **data quality, consistency, and better model performance.**

🔧**Preprocessing Steps:**

- Identified and handled **missing or inconsistent values**
- Applied **Label Encoding & One-Hot Encoding for categorical features**
- Performed **feature scaling using StandardScaler**
- Conducted **feature selection** to remove **irrelevant/redundant attributes**
- Split dataset into **training (80%)** and **testing (20%) sets**
- These steps help in **improving accuracy, reducing overfitting,** and ensuring **efficient model learning.**

Exploratory Data Analysis

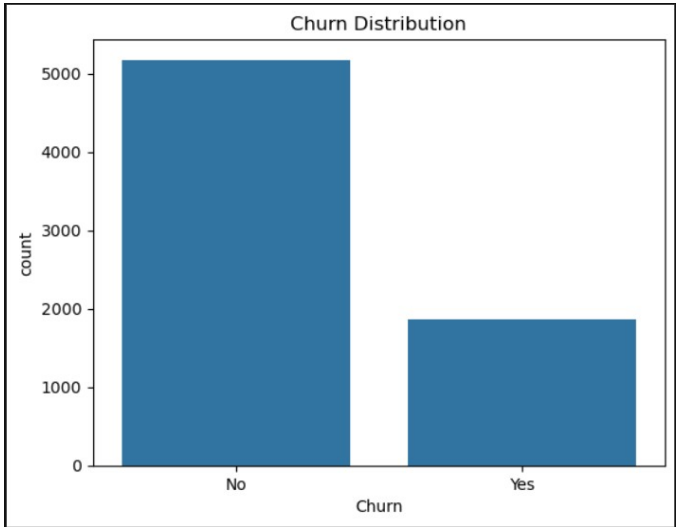
EDA was conducted to **uncover patterns, trends, and relationships** within the dataset.

📊 **Key Insights:**

- Customers using **Fiber Optic internet services** show higher churn rates compared to DSL users
- Contract type plays a major role — **month-to-month customers** are more likely to churn
- Long-term contracts (**one-year / two-year**) show **lower churn probability**
- Distribution of **contract types** shows majority customers prefer flexible plans
- These insights highlight how **service type and contract duration** influence churn behaviour
- **Fiber optic users** tend to have higher churn rates

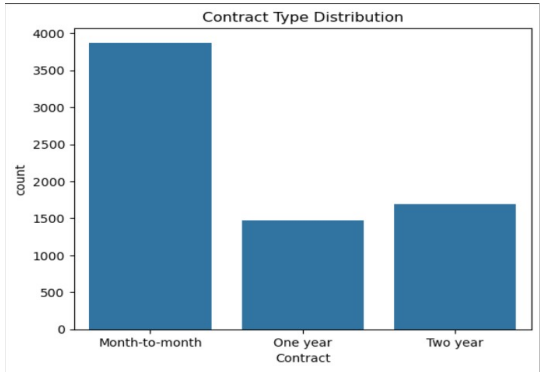
📈 Techniques Used:

- Data visualisation (bar charts, count plots)
- Correlation analysis
- Comparative feature analysis
- EDA plays a vital role in **feature selection** and **model improvement.**



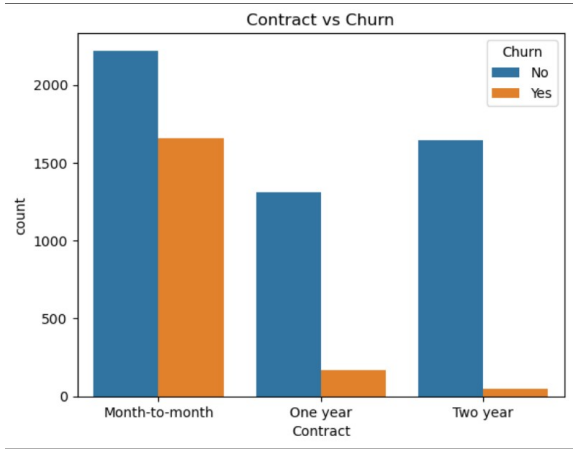
Observation

The graph shows that the number of customers who did not churn is higher than those who churned. This indicates that the dataset is imbalanced, with more customers staying than leaving.



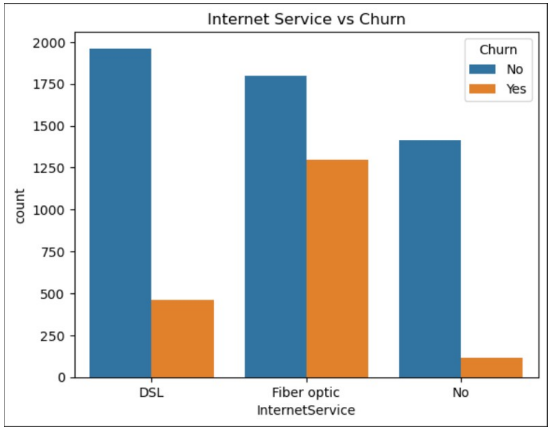
Observation

Different contract types are present, with month-to-month being the most common.



Observation

Month-to-month contract customers show higher churn.



Observation

Customers using Fiber Optic internet appear to churn more compared to DSL users.

Model Building

AI and ML models were developed to **predict customer churn effectively.**

⚙️ **Process:**

- Dataset split into **training and testing subsets**
- Applied **Logistic Regression** as a **baseline model**
- Implemented **XGBoost Classifier** for **enhanced performance**
- Trained models on **processed dataset**
- Performed **hyperparameter tuning**
- Saved final model using **Pickle (.pkl)**

🤖 **Model Justification:**

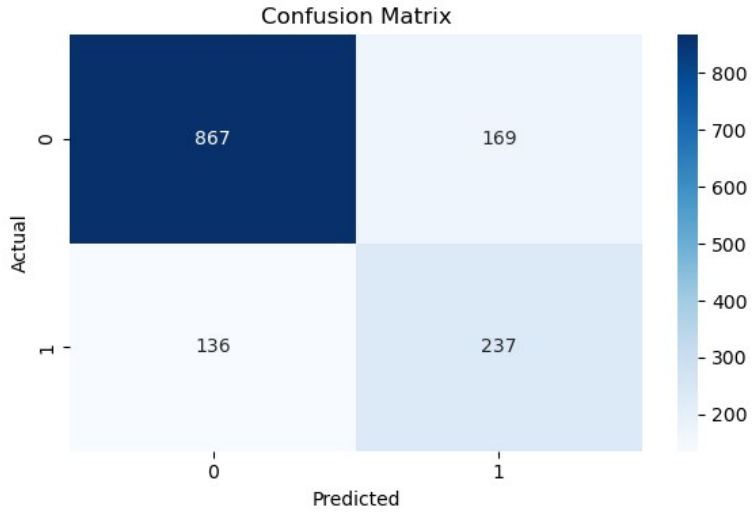
- **Logistic Regression** → Simple, interpretable, efficient
- **XGBoost** → Handles **complex patterns,** improves **accuracy**
- Ensures **robust and reliable** predictions.

Model Evaluation

- Model performance was evaluated using **multiple metrics** to ensure **accuracy and reliability.**

  📊 **Evaluation Metrics:**

- **Accuracy Score** → Overall correctness
- **Precision & Recall** → Performance on churn class
- **F1-Score** → Balance between precision & recall
- **Confusion Matrix** → Detailed classification results
- **ROC Curve & AUC Score** → Model's discrimination ability
- Ensures model performs well on **unseen data** and avoids bias.

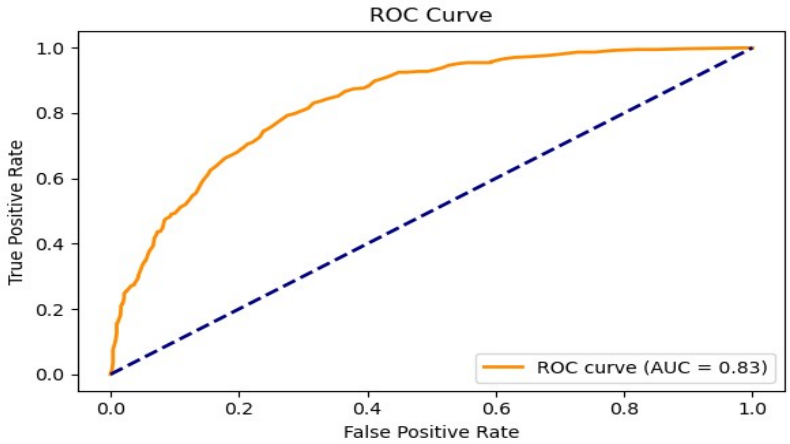


The Confusion Matrix shows the classification performance of the model by comparing actual vs predicted values.

It highlights how accurately the model identifies churn and non-churn customers, including correct predictions and errors.

The ROC Curve represents the model's ability to distinguish between churn and non-churn customers.

A higher **AUC score** indicates better model performance and strong classification capability.



Results

- The model achieved **strong predictive performance**
- Successfully identified **high-risk churn customers**
- Provided **actionable business insights**
- Demonstrated effectiveness of **machine learning & AI in real-world scenarios**
- Supports **proactive customer retention strategies**
- Successfully **identified customers at high risk of churn**
- The results highlight the **importance of data-driven decision-making** in **customer retention.**

Deployment

- The trained model was deployed using **Streamlit**, enabling **real-time interaction.**

  💻 **Application Features:**

- **Intuitive and user-friendly interface**
- Accepts **customer input parameters**
- Provides **instant churn prediction**
- Can be used by **non-technical stakeholders**
- This deployment makes the model **practical and accessible** for real-world use.

Conclusion

- Developed a **robust churn prediction system**
- Successfully applied **machine learning techniques**
- Identified **key factors influencing churn**
- Demonstrated **end-to-end ML pipeline**
- Shows how **predictive analytics** can improve **business growth and retention.**

Future Scope

- Improve performance using **advanced models** (Random Forest, Deep Learning)
- Integrate **real-time and streaming data**
- Deploy on **cloud platforms** (AWS)
- Enhance **UI/UX and scalability**
- Use **AI-driven personalisation for customer retention**
- To integrate **real-time prediction systems** using **streaming data pipelines** for continuous churn monitoring
- To implement **personalised recommendation systems** using **AI/ML** to improve **customer engagement and retention**

THANK YOU..